

Problem

A step-down power transformer with a turns ratio of $n = 0.1$ supplies 12.6 V rms to a resistive load. If the primary current is 2.5 A rms, how much power is delivered to the load?

Step-by-step solution

Step 1 of 1

$$n = \frac{V_2}{V_1} = \frac{I_1}{I_2}$$

Or,

$$\begin{aligned} I_2 &= \frac{I_1}{n} \\ &= \frac{2.5}{0.1} = 25 \text{ A} \end{aligned}$$

$$\begin{aligned} P &= IV \\ &= 25 \times 12.6 \\ &= 315 \text{ Watts} . \end{aligned}$$